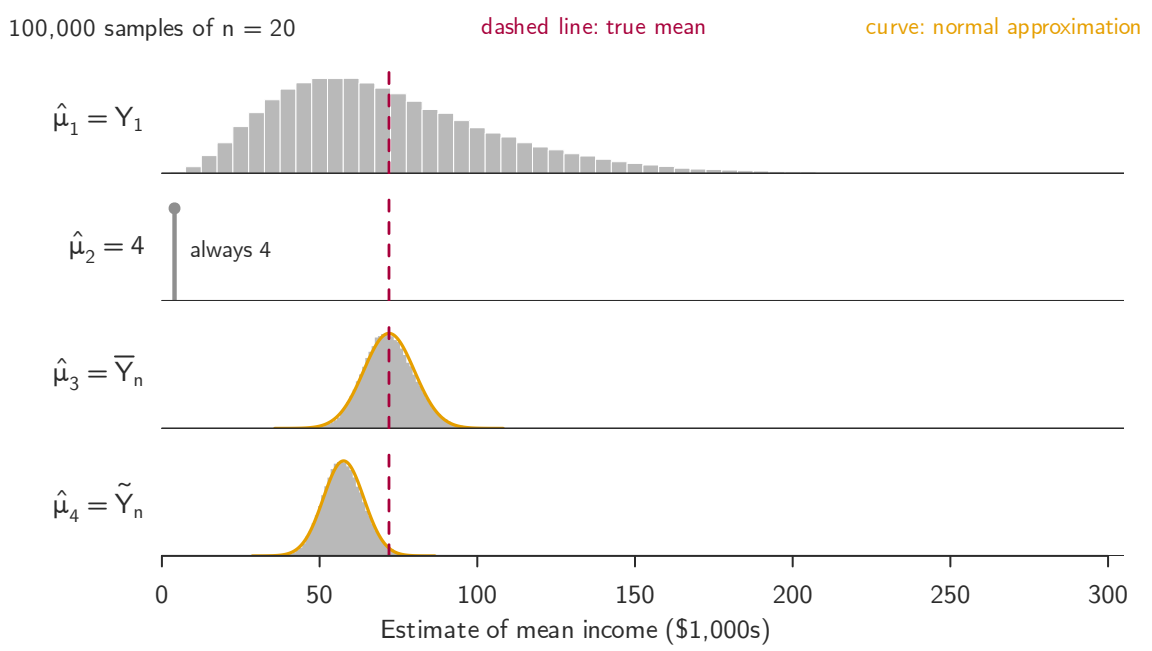
100,000 samples of n = 20
dashed line: true mean
curve: normal approximation
$\hat{\mu}_1 = Y_1$
$\hat{\mu}_2 = 4$
always 4
$\hat{\mu}_3 = \overline{Y}_n$
$\hat{\mu}_4 = \widetilde{Y}_n$
0
50
100
150
200
250
300
Estimate of mean income ($1,000s)